

RNase cleavage products in mzLib

Succinct biological reference and implementation-validation report

Purpose. Establish the cleavage context, terminal chemistry, and evidence needed to validate the RNase definitions and digestion products in the current mzLib clone.

Scope. Part I contains only RNases with reported single-stranded RNA cleavage activity. RNase 1 (mammalian) and RNase I (E. coli) are separate enzymes. Condition-specific PhyM rows are retained. Literature-only RNases are included for validation coverage; they are marked *Not modeled* in the mzLib column.

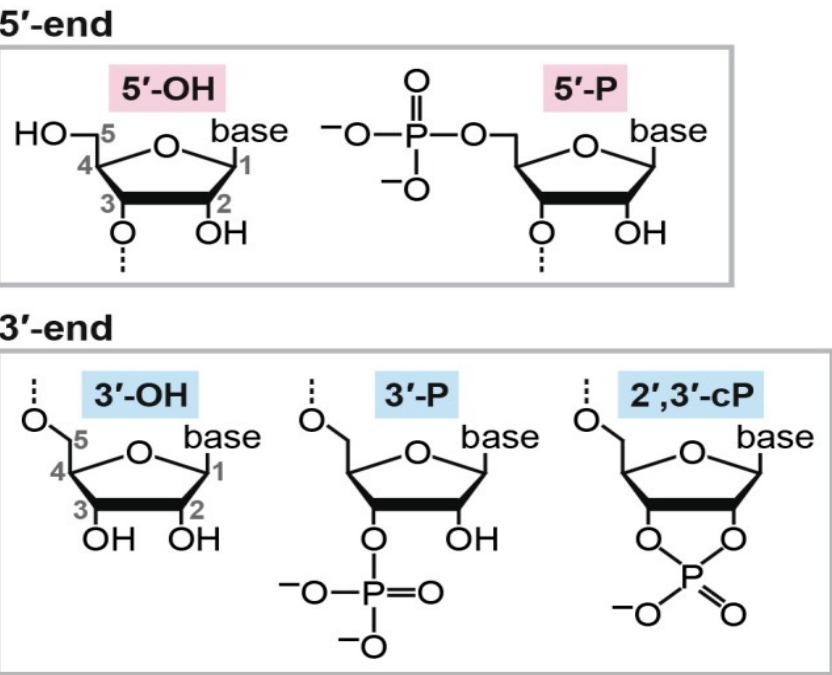


Figure 1. 5' and 3' RNA terminus structures used for terminology and formula mapping. Source: PMC6277466, Figure 1 (supplied image).

Formula key

Column	Formula	Meaning at a newly created terminus	Implementation/source
3' product	H2O4P	Linear 3'-phosphate remainder	Rnase.DefaultThreePrimeTerminus; E001
3' product	O3P	2',3'-cyclic-phosphate remainder	Existing cyclic-phosphate modification; E004,E029
5' product	O-3P-1	5'-OH remainder (removes phosphate)	Rnase.DefaultFivePrimeTerminus; E001
blank	-	Parser applies the corresponding Rnase default; this is not a measured zero	E001,E003

For an internal cut, the upstream fragment receives the 3' product and the downstream fragment receives the 5' product. The original RNA termini are retained when a product includes the source end.

Part I. Biological cleavage and products

Percentages are study-specific efficiencies, not universal enzyme rate constants. Where a source reports only preference, spacing, or structure dependence, that is stated instead of converted into a percentage.

RNase	Biological cleavage	mzLib cleavage	3' biological product	3' formula	5' biological product	5' formula	Occurrence / frequency evidence	Sources
RNase T1	Cleaves after G in ssRNA (GpN).	G (full)	3'-P via cyclic intermediate	H2O4P	5'-OH	O-3P-1	G sites observed throughout defined test RNA under optimal digestion; no missed cuts reported.	E006,E015
RNase T2	ssRNA endonuclease; little base specificity.	G ,C ,A ,U (full)	3'-P via cyclic intermediate	H2O4P	5'-OH	O-3P-1	Single-strand specific; no strong base preference; no universal rate reported.	E007,E025
RNase A	Cleaves after unpaired C or U.	C ,U (full)	3'-P via cyclic intermediate	H2O4P	5'-OH	O-3P-1	C and U sites; C reported about 2x faster than U in one summary; structure and conditions also matter.	E008,E028
RNase 1	Mammalian pancreatic-type RNase; ssRNA cleavage 3' to pyrimidines.	G ,C ,A ,U (full)	3'-P via cyclic intermediate	H2O4P	5'-OH	O-3P-1	No universal site rate reported; mammalian RNase 1 is a distinct enzyme from E. coli RNase I.	E030,E031
RNase I	E. coli endoribonuclease; largely sequence-independent on ssRNA.	G ,C ,A ,U (full)	3'-P via cyclic intermediate	H2O4P	5'-OH	O-3P-1	Broad ssRNA cleavage; no universal site rate reported. Ca2+ can support dsRNA activity, outside this ssRNA scope.	E009
RNase PhyM (7 M urea)	Cleaves Up-N and Ap-N under 7 M urea, 50 C, pH 5.	A ,U (full)	2',3'-cyclic-P observed	O3P	Not reported		Almost exclusive and uniform at Up-N/Ap-N under stated sequencing conditions; no percentage reported.	E010,E029
RNase PhyM (no urea)	Cleaves Up-N, Ap-N, and Gp-N without urea.	A ,U ,G (full)	2',3'-cyclic-P observed	O3P	Not reported		Gp-N is additionally attacked without urea; relative efficiencies not reported.	E010,E029
RNase 4	Cleaves after U in U-A and U-G ssRNA sites.	U A,U G (full)	Mixed: linear 3'-P and 2',3'-cyclic-P	H2O4P,O3P	5'-OH	O-3P-1	UA is slightly preferred over UG; no absolute site percentage reported.	E011,E018
RNase U2	Purine-selective cleavage after A or G; A preferred.	G ,A (full)	3'-P via cyclic intermediate	H2O4P	5'-OH	O-3P-1	A > G >> C > U reported; no universal rate.	E012,E013
Cusativin	Cleaves primarily at Cp[A/U/G] and UpA; additional UpU/ApU activity is reported.	C (full)	Mixed: linear 3'-P and 2',3'-cyclic-P	H2O4P,O3P	5'-OH	O-3P-1	CpA 70-90%; CpG/CpU 80-100%; UpA 70-90%; UpU/ApU 14-30%; CpC 0% in the LC-MS study. Waters identifies Cp[A/U/G] and UpA as primary sites.	E034,E015
RNase MC1	Cleaves at the 5' side of U; ApU/CpU/UpU preferred, GpU nearly resistant.	U (full)	Mixed: linear 3'-P and 2',3'-cyclic-P	H2O4P,O3P	5'-OH	O-3P-1	ApU 70-100%; CpU 80-100%; UpU 45-60%; GpU 1.0-1.5%; CpA 10-20%.	E015
colicin E5	Sequence/structure-specific tRNase at anticodon positions 34-35; Q-U/G-U context.	G U (full)	2',3'-cyclic-P	O3P	5'-OH	O-3P-1	Cleaves susceptible tRNAs at positions 34-35; not transcript-wide.	E016
RNase CL3	Chicken-liver cytidine-preferential ssRNA endoribonuclease.	Not modeled	Not established		Not established		C-containing to U-containing bond discrimination was 61:1 in limit digests of 5S/5.8S rRNA.	E024
RNase L	2-5A-activated, metal-independent ssRNA endoribonuclease; predominantly after U.	Not modeled	2',3'-cyclic-P	O3P	5'-OH	O-3P-1	Preference approximately UU > UA >> UG > UC; magnitude depends strongly on local structure.	E019,E026
RNase E	Low-specificity cleavage in AU-rich ssRNA; 5'-monophosphate-assisted pathway.	Not modeled	Not established		Not established		Prefers >=3 unpaired 5' nucleotides; optimal spacing about 8 nt; U/A/G-rich constructs diffused 2-3x faster than C-rich RNA in one assay.	E023,E027
RNase G	RNase E homolog; low-specificity cleavage in AU-rich ssRNA.	Not modeled	Not established		Not established		Prefers >=3 unpaired 5' nucleotides; optimal spacing about 6 nt; more impeded than RNase E with <4 unpaired nucleotides.	E023

Biological product chemistry is not known for every enzyme. A blank formula in the implementation table means the reader uses the configured default; a blank biological-product field means the cited literature did not establish that terminus.

Part II. mzLib implementation and resulting products

The following block is generated directly from the current clone's embedded rnases.tsv. It is retained as a text-format comparison against Part I.

```
Name | Motif | Specificity | ThreePrimeTerminusRemainder | FivePrimeTerminusRemainder | Notes
RNase T1 | G| | full | H2O4P | O-3P-1 | G, ssRNA
RNase T2 | G|,C|,A|,U| | full | H2O4P | O-3P-1 | ssRNA
RNase A | C|,U| | full | H2O4P | O-3P-1 | ssRNA
top-down | | none
RNase 1 | G|,C|,A|,U| | full | H2O4P | O-3P-1 | Mammalian RNase 1, slight pyrimidine preference
RNase PhyM (>= 7M urea) | A|,U| | full | O3P | | Up-N/Ap-N; almost exclusive and uniform under 7 M urea
RNase PhyM (< 7M urea) | A|,U|,G| | full | O3P | | Up-N/Ap-N/Gp-N
RNase 4 | U|A,U|G | full | H2O4P,O3P | O-3P-1 | U-A slightly preferred over U-G; ssRNA; mixed 3' products
RNase U2 | G|,A| | full | H2O4P | O-3P-1 | Purines, A preferred over G
Cusativin | C[C]|,U|A,X|U | full | H2O4P,O3P | O-3P-1 | Primary sites Cp[A/U/G] and UpA; CpA 70-90%; CpG/CpU 80-100%; CpC resistant; mixed 3' products; Waters 720008488
RNase_MC1 | [G]|U | full | H2O4P,O3P | O-3P-1 | ApU 70-100%; CpU 80-100%; UpU 45-60%; GpU 1.0-1.5%; mixed 3' products
colicin_E5 | G|U | full | O3P | O-3P-1 | Sequence/structure-specific tRNase at anticodon positions 34-35
RNase I | G|,C|,A|,U| | full | H2O4P | O-3P-1 | E. coli RNase I; largely sequence-independent ssRNA endonuclease
non-specific | X| | full | H2O4P | O-3P-1 |
singleN | | SingleN | H2O4P | O-3P-1 |
singleC | | SingleC | H2O4P | O-3P-1 |
```

Implementation checks

The | markers above render the tab boundaries for readability; the six-column order is unchanged.

Check	Current behavior	Evidence
Formula defaults	Omitted formula fields become singleton lists: 3' H2O4P and 5' O-3P-1.	Rnase.cs lines 9-21; E001
List-valued termini	ThreePrimeTerminusRemainder and FivePrimeTerminusRemainder are IList values.	Rnase.cs lines 12-21; E001
TSV parsing	RnaseDictionary resolves both formula columns, splits comma-separated formulas, and constructs Rnase with both lists.	RnaseDictionary.cs lines 293-397; E003
Multiple products	Internal products enumerate the Cartesian product of the 3' and 5' remainder lists.	Rnase.cs GetDigestedTermini; E001
RNase 4	H2O4P,O3P encodes two alternative 3' products; this represents alternatives, not probabilities.	rnases.tsv; E011,E018
PhyM	Two condition-specific rows: A/U at >=7 M urea and A/U/G below 7 M urea. The blank 5' field therefore uses the default in code, although the biological 5' terminus is unreported.	rnases.tsv; E010,E029
Backward compatibility	Existing three-column files remain valid because formula columns are optional and constructor defaults preserve prior behavior.	RnaseDictionary.cs; E001,E003

Worked mapping. For an internal cut in A-C|G-U, the default expected pair is upstream A-C with 3' remainder H2O4P and downstream G-U with 5' remainder O-3P-1. If the RNase supplies O3P as an alternative, the upstream product has both formula alternatives. Values on both sides produce all pairwise combinations.

Validation state and limitations

Source snapshot. Current clone commit: 65cb60f70867125d496b7cf5655a3ddbfff978a8. The report is generated from rnases.tsv, not from the older E: worktree. The supplied terminus figure is embedded in the PDF.

Validation gate. Scientific source verification remains a human responsibility: URLs and locators below provide traceability, but this report does not certify that every external source has been independently re-opened.

Local regression. The transcriptomics-filtered test run in this clone completed 449/449 tests with zero failures on 2026-09-15; the build emitted existing nullable-reference warnings.

Limitations. Motif syntax does not encode secondary structure, quantitative cleavage weighting, enzyme activation, or mixed-product distributions. Literature-only RNases require separate modeling decisions. For PhyM, the cited literature supports condition-specific specificity and a 3' cyclic-phosphate assignment, but not a numeric efficiency or biological 5' terminus.

References and provenance

ID	Source	Locator	URL / location
E001	mzLib Rnase.cs (local implementation).	Lines 9-21, 73-255.	local
E002	mzLib rnases.tsv (embedded RNase definition table).	Lines 1-112.	local
E003	mzLib RnaseDictionary.cs (active TSV parser).	Lines 42-75, 293-397.	local
E004	mzLib RnaMods.txt (local terminal modifications).	Remainder-modification entries.	local
E005	mzLib transcriptomics tests (local validation).	Test/Transcriptomics/.	local
E006	ENZYME EC 4.6.1.24: ribonuclease T1.	Reaction and comments.	https://enzyme.expasy.org/EC/4.6.1.24
E007	ENZYME EC 4.6.1.19: ribonuclease T2.	Reaction and comments.	https://enzyme.expasy.org/EC/4.6.1.19
E008	Thermo Fisher Scientific: Deciphering Nuclease Digestion Data.	RNase A/T1 specificity discussion.	https://www.thermofisher.com/us/en/home/references/ambion-tech-support/rn-a-structure-function/tech-notes/deciphering-nuclease-digestion-data.html
E009	E. coli RNase I exhibits a strong Ca2+-dependent inherent dsRNA activity (2021).	Abstract and introduction.	https://doi.org/10.1093/nar/gkab284
E010	Donis-Keller (1980), Phy M: an RNase activity specific for U and A residues.	Abstract.	https://pmc.ncbi.nlm.nih.gov/articles/PMC324360/
E011	NEB M1284: RNase 4.	Product description.	https://www.neb.com/en-us/products/m1284-rnase-4
E012	Improving RNA modification mapping by LC-MS through RNase U2-E49A (2018).	Introduction.	https://pmc.ncbi.nlm.nih.gov/articles/PMC6214470/
E013	ENZYME EC 4.6.1.20: ribonuclease U2.	Reaction and comments.	https://enzyme.expasy.org/EC/4.6.1.20
E015	RNA Cleavage Properties of RNase MC1 and Cusativin (2022).	Abstract and Results 2.1.1.	https://pmc.ncbi.nlm.nih.gov/articles/PMC9266746/
E016	Structural basis for sequence-dependent recognition of colicin E5 tRNase (2006).	Abstract and cleavage-product discussion.	https://pmc.ncbi.nlm.nih.gov/articles/PMC1669751/
E018	Human RNase 4 improves mRNA sequence characterization by LC-MS/MS (2022).	Figure 1D and UA/UG discussion.	https://academic.oup.com/nar/article/50/18/e106/6649373
E019	Ribonuclease L and metal-ion-independent endoribonuclease cleavage sites (2014).	Abstract, introduction, Table 1.	https://pmc.ncbi.nlm.nih.gov/articles/PMC4005677/
E023	Distinct requirements for 5'-monophosphate-assisted cleavage by RNase E/G (2016).	Abstract and introduction.	https://pmc.ncbi.nlm.nih.gov/articles/PMC4777840/
E024	Purification and properties of chicken liver RNase CL3 (1980).	Abstract: 61:1 C/U discrimination.	https://pubmed.ncbi.nlm.nih.gov/6243659/
E025	Several RNase T2 enzymes function in Tetrahymena RNA turnover (2012).	Abstract: ssRNA specificity/base preference.	https://pmc.ncbi.nlm.nih.gov/articles/PMC3248902/
E026	New advances in understanding RNase L (2020).	RNase L cleavage preference.	https://pmc.ncbi.nlm.nih.gov/articles/PMC7128181/
E027	RNase E searches for cleavage sites by linear diffusion (2024).	Abstract/results: sequence-dependent rates.	https://academic.oup.com/nar/article/52/11/6674/7655786
E028	RNase A: Frequently Asked Questions.	Reported C-versus-U rate difference.	https://agscientific.com/blog/rnase-a-faq.html
E029	Sequencing RNA by exonuclease digestion and uridine-specific cleavage using MALDI-TOF (1998).	PhyM product assignment.	https://academic.oup.com/nar/article/26/2/446/1170657
E030	Nucleotide substrate binding characterization in human pancreatic-type ribonucleases (2019).	Human RNase substrate preferences.	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0220037
E031	Pancreatic Ribonuclease.	RNASE1 entry and occurrence.	https://pancreapedia.org/molecules/pancreatic-ribonuclease
E032	PMC6277466, supplied terminus-structure figure.	Figure 1; image reproduced for terminology key.	https://pmc.ncbi.nlm.nih.gov/articles/PMC6277466/
E034	Waters RapiZyme MC1 and Cusativin Care and Use Manual, Lit. 720008488.	Figure 1 and Cusativin specificity discussion: primary Cp[A/U/G] and UpA sites.	https://help.waters.com/help/en/support/library-details.html?documentid=720008488 (PDF: https://help.waters.com/content/dam/waters/en/support/usermanuals/2024/720008488/720008488.pdf)
E033	Kassis et al. (2026), Scientific Agent Skills.	Procedural source for evidence binding and audit structure.	https://doi.org/10.48550/arXiv.2609.00065

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